Technology Stack Details

Technology selection and justification

Project: AI WeatherWise API   Track: AI-ML & GEN-AI Track

Prepared: 07 October 2026   Team ID: Not provided

Maximum Marks: 2

Team Members & Assigned Roles

Team Member	Role
Jai Bharath A	Team Lead & Lead Backend/API Architect
Mukesh Balaji	Generative AI & Prompt Engineering Engineer
Yadagiri Mani	Database & Data Engineering Specialist
Deepak G	Security, Testing & Reliability Engineer
Naveen	API Integration, Documentation & Demonstration Engineer

Layer	Technology	Purpose / Justification
Runtime	Node.js	Async I/O and fast REST backend development.
Framework	Express.js	Routing, middleware, controller orchestration and REST API implementation.
Database	MongoDB Atlas	Cloud document storage suitable for flexible user/location records.
ODM	Mongoose	Schema validation, models, hooks and indexes.
Weather API	OpenWeatherMap	Real-time meteorological observations.
Gen-AI	Google Gemini 2.5 Flash	Natural-language summaries and contextual recommendations.
AI SDK	@google/generative-ai	Programmatic access to Gemini.
Security	bcryptjs + jsonwebtoken	Password hashing and 30-day Bearer authentication.
Testing	Thunder Client / Postman Collection	Endpoint regression, status and response validation.
Version control	Git + GitHub	Source history, collaboration and submission.
Deployment path	Docker / Render / cloud-ready	Reproducible packaging and future deployment.

Architecture Selection Rationale

- Node.js + Express keeps the API lightweight and modular.
- MongoDB fits the nested/variable nature of weather-related metadata and favorite-location notes.
- Gemini adds the generative layer; deterministic rules remain the safety net.
- JWT allows stateless authentication across clients.
- Environment variables prevent secrets from being hard-coded.

Configuration Variables

Variable	Purpose
PORT	HTTP server port
MONGODB_URI	MongoDB Atlas connection string
JWT_SECRET	JWT signing secret
OPENWEATHER_API_KEY	OpenWeatherMap credential
GEMINI_API_KEY	Gemini API credential